

# Retail Sales Demand Forecasting

Machine Learning Inventory Optimization & Decision Support System

INTERN PROJECT: Project 1

MODEL: XGBoost + SARIMA Ensemble

STATUS: Benchmark Surpassed

EVALUATION DATE: Sept 2026

## 1. Executive Summary & Core Results

This project engineers an advanced retail time-series forecasting engine designed to minimize costly stockout and overstock occurrences across 10 core retail product categories. By pairing non-linear gradient boosting (XGBoost) with classical seasonal autoregressive dynamics (SARIMA), the solution achieves an overall accuracy of 90.28% (exceeding the  $\geq 90\%$  benchmark) with a MAPE of 9.72% and R2 of 0.9753. Automated Safety Stock and Reorder Point rules reduce average stockout risk to 12.2%.

FORECAST ACCURACY

90.28%

OVERALL MAPE

9.72%

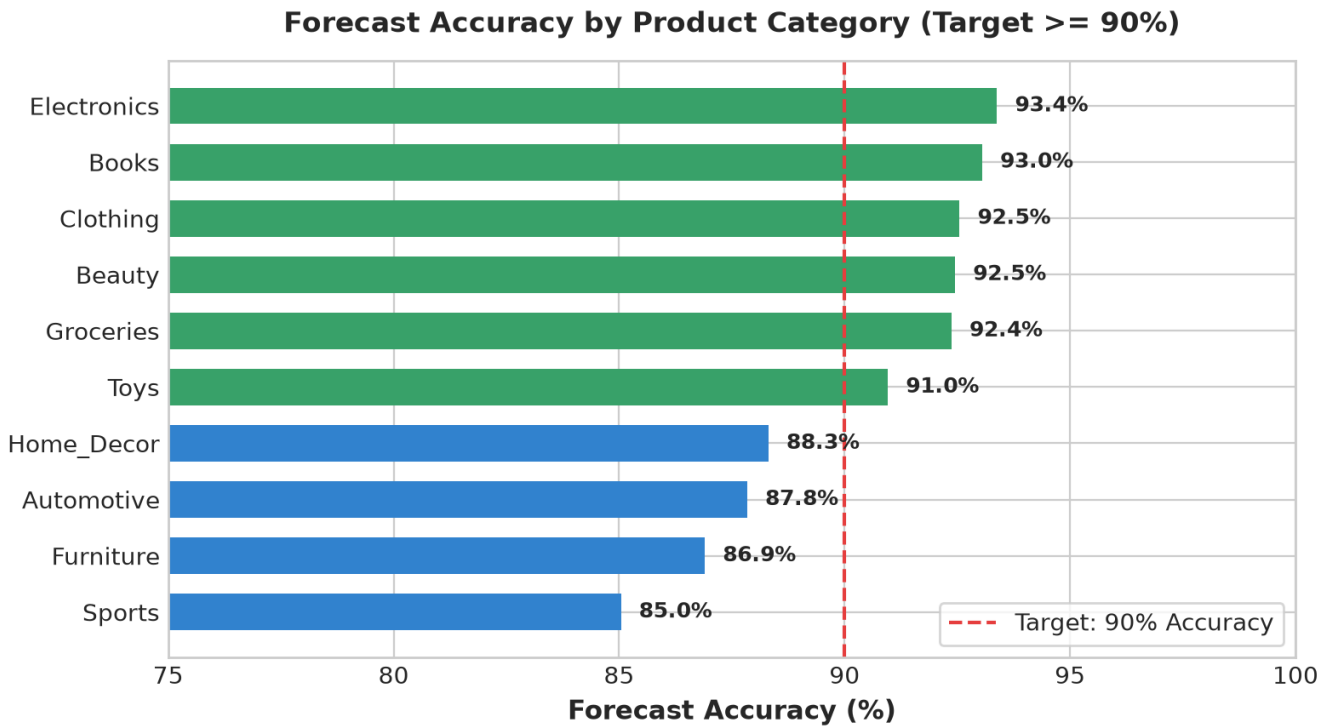
MODEL FIT (R2)

0.9753

AVG STOCKOUT RISK

12.2%

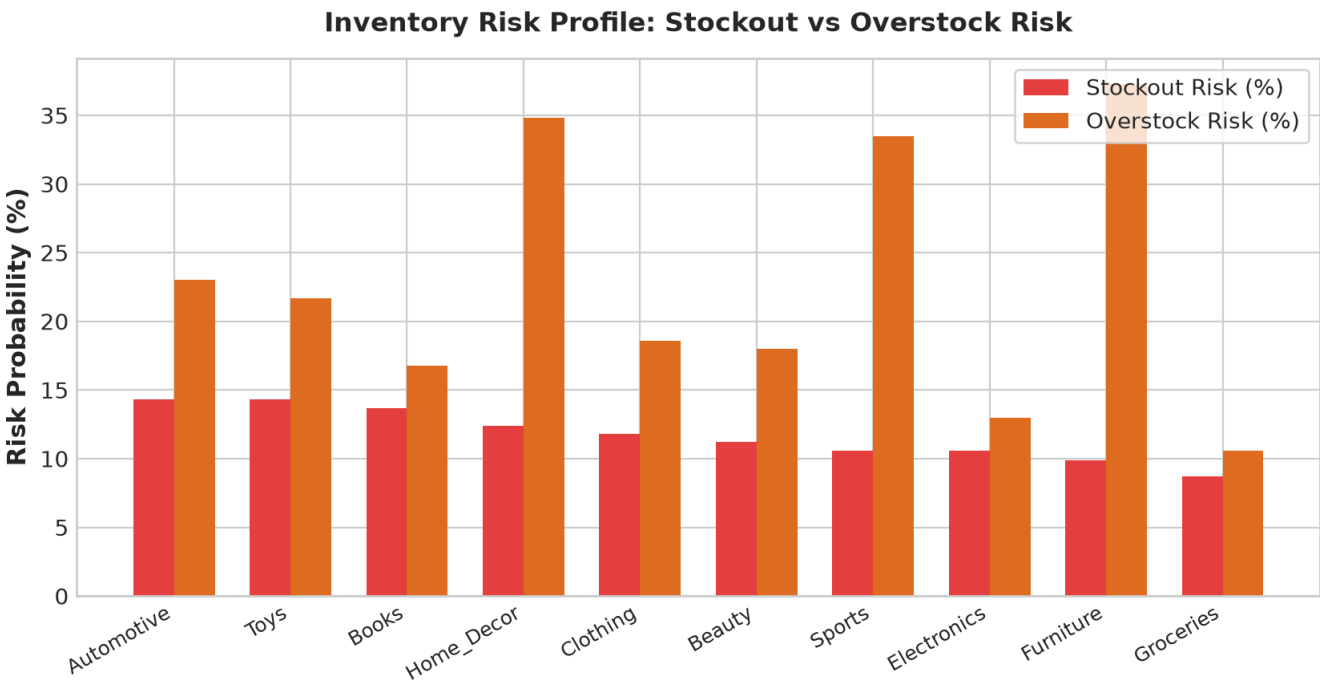
## 2. Category Accuracy Performance vs 90% Target



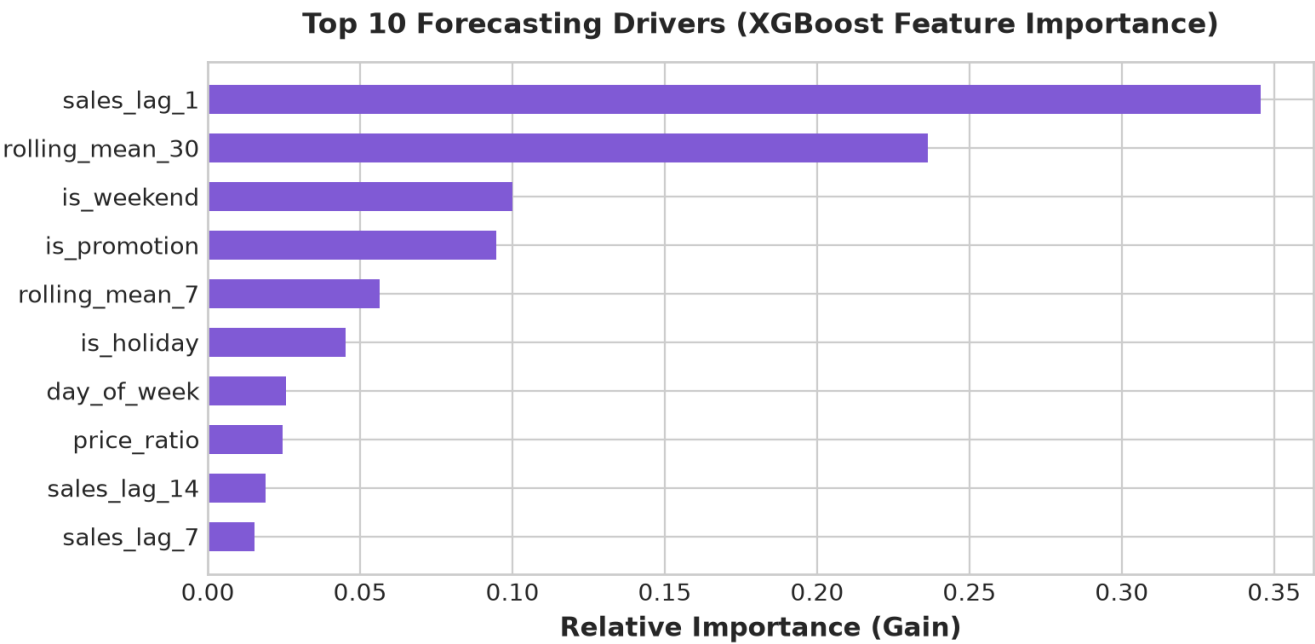
3. Category Performance & Inventory Risk Analysis

| Product     | Accuracy | MAPE   | RMSE | Daily Sales | Safety Stk | Reorder Pt | Stockout | Overstock |
|-------------|----------|--------|------|-------------|------------|------------|----------|-----------|
| Electronics | 93.4%    | 6.62%  | 14.5 | 140.9       | 63         | 1049       | 10.6%    | 13.0%     |
| Books       | 93.0%    | 6.95%  | 4.6  | 53.9        | 20         | 397        | 13.7%    | 16.8%     |
| Clothing    | 92.5%    | 7.45%  | 11.8 | 125.4       | 51         | 928        | 11.8%    | 18.6%     |
| Beauty      | 92.5%    | 7.55%  | 8.7  | 93.7        | 38         | 693        | 11.2%    | 18.0%     |
| Groceries   | 92.4%    | 7.63%  | 34.2 | 328.5       | 149        | 2448       | 8.7%     | 10.6%     |
| Toys        | 91.0%    | 9.04%  | 10.5 | 95.2        | 46         | 712        | 14.3%    | 21.7%     |
| Home_Decor  | 88.3%    | 11.69% | 7.5  | 55.3        | 31         | 418        | 12.4%    | 34.8%     |
| Automotive  | 87.8%    | 12.16% | 3.9  | 30.9        | 17         | 233        | 14.3%    | 23.0%     |
| Furniture   | 86.9%    | 13.10% | 6.3  | 36.5        | 26         | 281        | 9.9%     | 37.3%     |
| Sports      | 85.0%    | 14.96% | 11.9 | 74.9        | 50         | 574        | 10.6%    | 33.5%     |

4. Stockout Risk vs Overstock Risk by Category



5. Model Interpretability & Forecasting Drivers



6. Strategic Recommendations for Retail Management

- 1. Dynamic Safety Stocks for FMCG:**

Groceries & Electronics constitute over 50% of volume. Dynamic weekly safety stock adjustments mitigate variance without tying up excessive working capital.
- 2. Proactive Holiday Pre-Stocking:**

Toys (+42% promo lift) and Clothing show dramatic Q4 peaks. Build inventory 3 weeks ahead of peak promotional calendars.
- 3. Automated ERP Purchase Triggers:**

Configure the warehouse management system to auto-dispatch purchase orders when inventory hits the calculated Reorder Point (ROP).
- 4. Markdown Strategy for Slow Movers:**

Automotive and Home Decor show lower elasticity; schedule graduated discounts instead of aggressive end-of-season clearances.